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Open Sesame: Inside a Data Thief's Treasure Trove

By Jordan Robertson

AP

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What kinds of information do hackers love to steal? Any and all. Whatever it is you're typing into a computer while online, a hacker may be able to put it to use for criminal endeavors. Their trick is to start small -- learn a few tidbits about a certain individual -- then work up to accessing much more sensitive information, up to and including an entire bank's computer system.

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Getting hacked is like having your computer turn traitor on you, spying on everything you do and shipping your secrets to identity thieves.

Victims don't see where their stolen data end up. But sometimes security 10th Annual Online Fraud Report. Online Payment Fraud Trends, Merchant Practices and Benchmarks. Get your copy today. researchers do, stumbling across stolen-data troves that offer a glimpse of what identity theft looks like from criminals' perspective.

Researchers from U.K.-based security firm Prevx found one such trove, a Web site used as a stash house for data from 160,000 infected computers before it was shut down this month.

The find offers a case study on just how much data criminals are stealing every day, from the utterly inconsequential to the alarmingly private.

It also shows the difficulty in shuttering criminals' ID-theft beachheads: The Web site that Prevx found, which was operating on a server in Ukraine, was still online for nearly a month after security researchers alerted the Internet service provider and law-enforcement authorities. The site was sucking up data from 5,000 newly infected computers each day.

Data Hidey Holes

The victims in the Prevx find are mostly everyday people handing over their passwords for Facebook More about Facebook and banking sites, along with their love notes and other e-mails. But more dangerous personal information is there, too, including Social Security numbers and other account information from one bank's infected computer.

Caches of stolen data like these are hidden throughout the Internet, usually locked away inside password-protected Web sites or heavily fortified servers. Prevx's researchers were able to infiltrate this site because it was protected with poor encryption.

In that sense, the find illustrates how even sloppy crooks can vacuum up enormous amounts of information through massive "botnets" -- armies of infected computers formed by spreading a computer virus More about computer virus that orders compromised machines to phone home for further instructions, such as sending out spam or relaying passwords.

The botnet Prevx found was only harvesting data, though Prevx said it could have been upgraded to do other things.

Watching Your Every Move

Ordinary Internet sessions are logged in great detail. One Southern California 22-year-old could be seen registering a domain name with GoDaddy.com, changing his Yahoo (Nasdaq: YHOO) More about Yahoo e-mail password and ordering a meal online from Pizza Hut. His credit card number, birth date, telephone number, address and passwords are now all in criminals' hands, though it's unclear what, if anything, criminals have done with the information yet.

Some victims are gold mines for sensitive data. An infected computer at a Georgia bank exposed customer details and credentials for the bank's wire-transfer system. Bank employees were checking e-mail, looking up BMWs and Infinitis and working with customers' accounts on the same infected machine.

Government computers were also hit, including one in Texas that coughed up Web site logins for one of the government's health care providers, and another in North Carolina that revealed access to an agency's human resources system.

"This is giving criminals the keys to the castle," said Prevx's director of malware research, Jacques Erasmus. "Once they're into this system, it might not seem at this point like it's the biggest data heist ever, but this is how they get into a network. This is their game -- they do this every day."

In other words, criminals start small, then use their first point of attack as a way to jump onto more sensitive computers.

Excuse Me, but You've Been Hacked

Researchers who discover these stolen-data caches then have to figure out what to do with them. Notifying victims is time-consuming and difficult, and researchers tend to focus on trying to get service providers to deactivate the servers before criminals get to the data on them.

Prevx said it alerted the site's Internet provider, the FBI and UK authorities about the breach it discovered. The company also talked to the affected bank, Doraville, Ga.-based Metro City Bank, a community bank whose Web site lists four locations, and Prevx said the bank has removed the infected computer.

One customer -- Yoon-Kee Hong, a 22-year-old college student from Suwanee, Ga. -- had signed up for an account with Metro City Bank just a month before learning about the breach. He said he had not been alerted by the bank that his Social Security number and other personal details were stolen.

After being told about the breach by The Associated Press, which picked his name from the files provided by Prevx, the student said he planned to cancel his account.

"I cannot trust them any more," he said. "They're not doing what they're supposed to

do. They didn't even notify me. It's like they're trying to hide it from their customers."

He later relented and decided to stay with the bank after he was offered a new account and promises of fraud alerts.

The bank said in a statement that it is notifying customers and is investigating the breach, refusing to comment further. State officials in North Carolina and Texas didn't return calls on the breaches there. The FBI didn't return a call about the breaches.

The Zombie Masters

Such finds are becoming more common as the barrier lowers for crooks to jump into the online identity-theft racket. Top-of-the-line viruses, also known as "Trojans," can be had for under \$1,000.

Joe Stewart, a SecureWorks botnet expert who was not involved in Prevx's research, said that last year, he helped shut down a command-and-control server for a huge botnet that had infected more than 378,000 machines and had stolen more than 460,000 usernames and passwords.

There are countless other smaller botnets, set up by less sophisticated criminals who steal as much data as they can and simply pull up stakes, and do it all over again, once their operation has been detected.

"The level of amateurism speaks to how widespread it is," Stewart said. "Literally anybody with a little bit of computer knowledge at all, if they have the criminal bent, can get access to one of these Trojans and get it out there and start stealing people's data."

Michael R Lord

COINS 315

9 September 2009

I4I's Word against Microsoft Word, an Ethical Overview

On August 11th 2009 a Texas Judge ordered that Microsoft discontinue the sales of the popular word processing tool Microsoft Word within 60 days and be fined over \$290 million (Gohring). The ruling was based on a copyright infringement upon software that was patented by the small Canadian company i4i. The fine is high, but the inability to sell America's most widely used word processing utility is the court's decision that everybody is watching. Unfortunately the details are not exactly crystal clear. Microsoft may have willfully infringed upon the copyright but the copyright itself is somewhat vague. If Microsoft is at fault, the punishments may be much more severe or far reaching than the judge had intended.

I4i patented a feature known as custom XML which allows people to make, "forms or templates such that words in certain fields are tagged and then can be managed in a database" (Gohring). Both Word 2003 and 2007 allow this feature despite the patent. I4i presented internal Microsoft e-mails that stated Microsoft would, "go ahead with the custom XML feature, even as they acknowledge the i4i patent". This caused the charges against Microsoft to become "willful infringement" charges and tacked on another \$40 million of damages (Keizer). Ultimately the judge decided to bar the sales of any version of Microsoft word that has the ability of opening an .XML, .DOCX or .DOCM file all of which contain custom XML (Robinson).

The Friday after the court ruling, Microsoft filed an "Emergency Motion" to continue the sales and shipping of Microsoft Word while an appeal takes place. Normally during an appeal only the monetary fines are suspended, but because Microsoft's real concern lies in the sales of Word, the emergency motion was necessary. This motion was approved by the US Court of Appeals on September 4th. Microsoft still believes that i4i's patents are invalid and is making preparations for its next court date of September 23rd (Protalinski). Although Microsoft was

granted this emergency motion, it is very possible that the sales of Microsoft word would have never stopped anyway.

The ability that Microsoft possesses to easily disable the XML feature would remove it from the Texas judge's ban. Barry Negrin, who has practices trademark and patent law for 17 years states that, "All Microsoft has to do is disable the custom XML feature, which should be pretty easy to do, then give that a different SKU number from what's been sold so it's easy to distinguish the two versions" (Keizer). This brings up an interesting situation where Microsoft can be fairly easily exempt from its sentence. The monetary fines are quite large, but to a company like Microsoft, they only hurt finances marginally. So while the punishments appear to be a death sentence to the word processing giant, it is only an inconvenience. Microsoft's employees know this, and it could be part of the reason for willingly infringing upon the patent. The scenario provides a situation where Microsoft can do almost whatever it pleases in an attempt to get ahead.

I4i is a company that is significantly smaller than Microsoft, and while Word is only a small portion of Microsoft's profit, custom XML is a relatively large portion of what i4i does. I4i claims that, "business shrank significantly when Microsoft added custom-XML support to Word" (Protalinski). This is extremely possible due to the fact that most businesses use Microsoft Office as a word processing suite. If a company already has the ability to create these custom XML documents, then they have no reason to contact i4i even if i4i holds the patent. Fortunately even though Microsoft may have known about the patent beforehand, the company is not providing a simple software fix to dodge the punishment. The emergency motion to ship Word was an ethical approach that, but the "actual document was filed under seal, so the full contents of the request have not yet been made public" (Albanesius).

It is possible that Microsoft argued the impact that the ban of selling Word would have on businesses. In an earlier court battle with eBay, the Supreme Court decided that lower courts need to use a four-factor test when conducting injunctions. One of these four factors is that “the public interest is not disserved with the injunction” (Keizer). Because Microsoft Word is so heavily used, it is about as public as something can get. While the injunction does not ban the use of software already purchased, the inability to purchase a new computer with Word on it or perhaps to purchase the software for education will impact the public negatively as well as other businesses. The judge may not have fully comprehended the impact that this sentence could have. Microsoft may have used this as an argument when it filed an emergency motion.

While the impact of a successful ban of Word would be far reaching, it is difficult to say that agencies would be hurt by it. There are other word processing options available that the average user is unaware of such as Open Office or online options like Google Docs. Ray Bjorklund, a senior vice president and chief knowledge officer at FedSources stated that, “Agencies might wait two or three years to refresh technologies. And those that need current versions of Word and Microsoft Office probably already have them” (Robinson). So a ban on Microsoft Word may not truly be as detrimental as some people would initially anticipate.

The events leading up to today have provided the possibilities of ethical wrongdoings by both Microsoft and the judge’s punishments. While the appeal, which could last over a year, is in progress it is best for those of us who are interested to keep an open mind by observing both sides of the issue. While the final verdict may not affect us directly as consumers, it could change the way that the courts deal with companies like Microsoft who are willingly committing copyright infringement, or ripple into changing the way that software is patented in the first place.

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Michael R Lord

COINS 315

February 3, 2009

Internet Gambling, Legal Win or Ethical Sin?

The internet has improved communication, improved the availability of information, and aided businesses in their pursuit of growth and wealth. One particular industry that found profitability through the internet is the industry of gambling. A report from the United States General Accounting Office estimated that internet gambling provided revenues of more than 4 billion dollars in the year 2003 alone (Internet). The easy access of the internet and the addictive nature of gambling are a toxic combination. Combined with the internet's lack of regulations online gambling becomes a dangerous addiction. Users are faced with new issues that were otherwise avoided such as lacking trust in the game they are playing, not knowing if the money they win will ever actually be given to them, and not knowing where the money they spend is actually going. These reasons, among others have caused our Government to enact both state and federal laws in an attempt to control the increasing risks of online gambling.

South Carolina's gambling laws are written specifically to regulate the traditional sense of gambling where people are physically present with the casino or hold physical tickets (South). The Government has previously decided to allow each state to regulate its own laws regarding gambling, but internet gambling allows someone in a state such as South Carolina to participate in gambling in another state (or country) right from the comfort of their own home. This presents a new problem for individual states that wish to tax gambling as an example. Because every state's laws are different, people will gravitate toward the website that offers the most profit with the least amount of taxes or regulations providing incredible profit for that one company and in turn state and potentially massive deficit for other states. As South Carolina's laws have not yet been adjusted to protect it from this sort of financial burden, the Federal Government has been creating new laws to prevent these issues among others.

In 2005 the Internet Gambling Prohibition and Enforcement act (H.R. 4411) was introduced. This bill was the first attempt to control internet gambling and it regulated it by banning gambling businesses from accepting payments such as credit cards, electronic funds transfers, and checks. This proved effective in limiting gambling to being physically present for the roll of the dice and passed the House floor by a bipartisan vote of 317 to 93. Currently, internet gambling has been included in Title VIII of the Safe Port Act and further clarifies the limitations and regulations placed against internet gambling, not only stateside, but overseas as well where U.S. law has no impact. Most internet gambling websites are actually hosted overseas so the only way for America to regulate it is to block electronic transfers of money to these sites. As the Safe Port Act was being voted upon, South Carolina representative Bob Inglis received stark criticism for his support.

A comment on Inglis' government website left by Justin Lively displays his frustration with the bill. "The issue is about much more than gambling. It's about government prohibiting activities it has no business prohibiting. Gambling is addictive? So is nicotine, alcohol, caffeine and sex...I simply can't vote for anyone who is continuing to bloat our Government with needless laws and further restrictions on our freedom (Inglis)." American citizens are against government regulations and some people view these regulations as limitations to their freedoms. While this freedom was not technologically available until recently, people feel that it is their right to spend their money as fruitfully or wistfully as they wish. Although the government has in the past left it up to individual states to decide gambling laws, it is simply not feasible in terms of the World Wide Web. Inglis claims that, "The ills of this form of unregulated gambling are manifold. Not only can it be highly addictive, but it fails to block access to children, and

provides tempting opportunities for organized crime.” Others point out that money profited from online gambling sites could be used for illegal activities such as money laundering.

Christians traditionally avoid gambling altogether and internet gambling should be no exception. Although the bible does not specifically prohibit gambling itself, it warns against a love of money (1 Timothy 6:10). The Bible describes money as one of the false idols that people tend to worship, making gambling increasingly addictive. Just as South Carolina’s laws were written before internet gambling, the bible was written before any slot machine was ever pulled, yet the guidelines still remain effective. From a Christian viewpoint, laws about internet gambling have no effect, but for the non Christians among us, these laws are encroaching on their freedoms.

One possible solution to the dilemma is to simply legalize internet gambling in the United States, where laws and regulations can easily ensure that the internet casino is fair, legal, and not supporting some sort of crime. Once again we are back to the initial problem of different state to state regulations and gambling across borders. The government could also outlaw gambling all together and start to develop a society that is far less dependent on money for entertainment. It is easy to see that most any solution to the complications of internet gambling is going to be costly to the state or federal government who is trying to enforce it, and no matter what path is chosen, people are going to feel as if they are losing another piece of their freedom.

Open borders inter-state and inter-continental gambling is potentially toxic for our economy and our society. Making addictive practices more easily available is not productive to anybody accept the owner of the website or casino and therefore should be avoided. Regardless

of the laws in place by either state or the federal government, it is much safer for citizens to take the Christian route and simply abstain from gambling altogether.

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Presentations

- **Presentation One**
 - Title: "Into Enemy Waters"
 - Class: Visual Basic Programming
 - Letter Grade Received : A

(Note: Presentations is part of a larger project grade)
- **Presentation Two**
 - Title: "The Foodinator"
 - Class: Applied Networking
 - Letter Grade Received : A

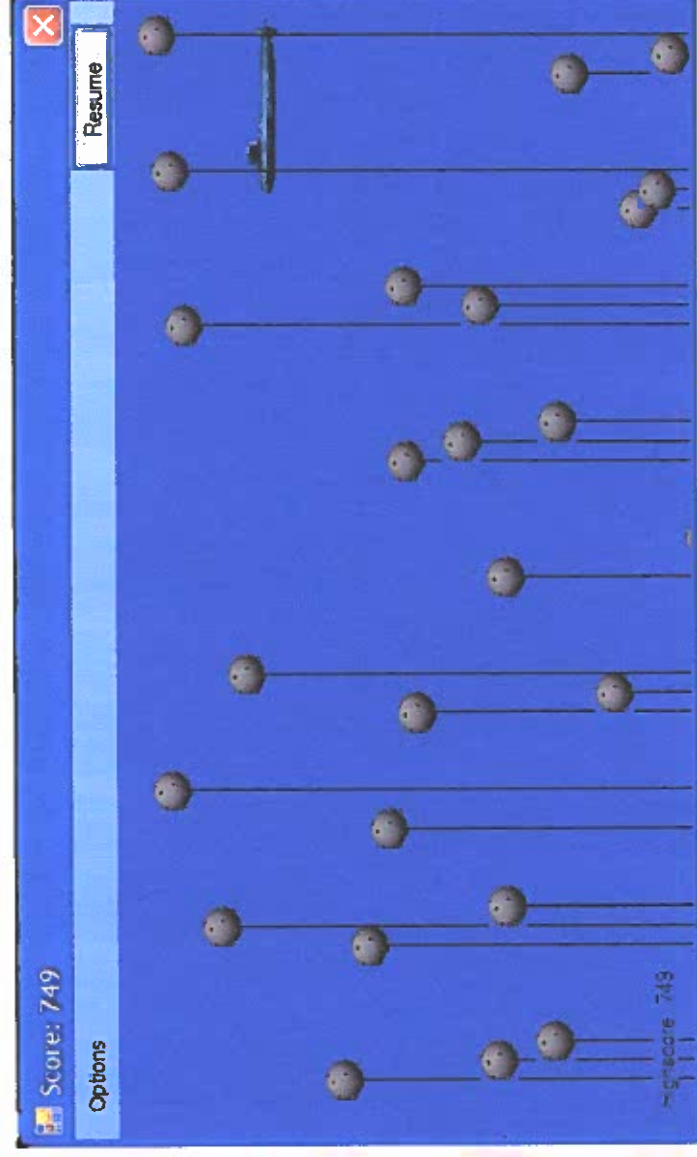
(Note: Presentations is part of a larger project grade)

Into Enemy Waters

Michael Lord

The Innovations

- ◆ The everything button
- ◆ Increasing difficulty
- ◆ Saved high scores



Everything Button

- ◆ Pauses, Resumes, and starts a new game!
- ◆ Simple Location



```
Private Sub Button1_Click(ByVal sender As System.Object, ByVal e As EventArgs) Handles Button1.Click
    If Button1.Text = "Pause" Then
        Button1.Text = "Resume"
        mytimer.Enabled = False
    ElseIf Button1.Text = "Resume" Then
        Button1.Text = "Pause"
        TextBox1.Visible = False
        mytimer.Enabled = True
    ElseIf Button1.Text = "New Game" Then
        TextBox1.Visible = False
        shipnum = 0
        clicktick = 0
        mclick = False
        Gameover = False
        shipnumtick = 0
        ReDim onscreen(0)
        onscreen(0) = False
        ReDim oby(0)
        ReDim obxpos(0)
        Gameoverlabel.Visible = False
        Button1.Text = "Pause"

        mytimer.Enabled = True
    End If
End Sub
```

Increasing difficulty

- ◆ Based on score
- ◆ All variables are arrays

```
Dim shipnumtick As Integer 'tracks score to add obstacle
Dim shipnum As Integer = 0 'tracks the number of obstacle on the screen
Dim onscreen() As Boolean = {False} ' detects weather or not obstacle is on the screen
Dim oby(0) As Integer 'obstacle y position
Dim obxpos(0) As Integer 'obstacle x position
```


Increasing difficulty (Cont)

- ◆ The for loops
- ◆ Why the number 23?

```
If (shipnumtick * 0.1) > 23 Then 'every time the user's score reaches 23... add another obstacle
    shipnum += 1
    shipnumtick = 0
    ReDim Preserve onscreen(shipnum)
    ReDim Preserve oby(shipnum)
    ReDim Preserve obxpos(shipnum)
    onscreen(shipnum) = False
End If
For I As Integer = 0 To shipnum
    If onscreen(I) = False Then ' testing whether each obstacle is on the screen
        oby(I) = Math.Floor(35 + randposition.NextDouble * 315)
        onscreen(I) = True
    End If
    If obxpos(I) > 700 Then
        onscreen(I) = False
        obxpos(I) = 0
    End If
    p.DrawImage(minebmp, New Point(obxpos(I) - 50, oby(I)))
    g.DrawLine(Pens.Black, (obxpos(I) - 40), (oby(I) + 20), (obxpos(I) - 40), Height)
Next
```

Saved high scores

- ◆ Save and exit
- ◆ Loading

```
Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles ExitToolStripMenuItem.Click
    Dim sw As IO.StreamWriter = IO.File.CreateText("../Highscore.txt")
    sw.WriteLine(highscore)
    sw.Close()
    Close()
End Sub
```

' saves highscore and close the program

```
Dim sr As IO.StreamReader = IO.File.OpenText("../Highscore.txt")
highscore = sr.ReadLine
sr.Close()
```


The Foodinator



**FINAL PRESENTATION
BY MIKE LORD**

What is the foodinator?



- An Online Database of recipes
- People can add their own recipes to the database
- People can search and view other recipes for a small fee

Features

- Different actions on recipe selection, based on account setup
- Adding your own recipe
- Unlimited or Single recipe account setups
- Recorded purchases for accounts set up as “Single” only.

Main Page (after login)

- A database grid layout acts as the method for searching/selecting a recipe.
- The select button activates a response to redirect you to the appropriate page based on your setup by using the SelectedIndexChanged event.

```
Protected Sub GridView1_SelectedIndexChanged(ByVal sender As Object, ByVal e As System.EventArgs)
    Dim row As GridViewRow = GridView1.SelectedRow
    Dim repID As String = row.Cells(13).Text
    Session("RecipeID") = repID
    Session("CurrentUser") = User.Identity.Name
    Response.Redirect("AgreeToBuy.aspx")
End Sub
```


Adding your own recipe

- Many validates to ensure proper input
- Block by block insert into the recipe database

```
Protected Sub Button1_Click(ByVal sender As Object, ByVal e As System.EventArgs) Handles Button1.Click
    If Page.IsValid Then
        SqlDataSource1.InsertParameters("RecipeName").DefaultValue = RecipeName.Text
        SqlDataSource1.InsertParameters("Category").DefaultValue = Category.Text
        SqlDataSource1.InsertParameters("Difficulty").DefaultValue = Difficulty.Text
        SqlDataSource1.InsertParameters("prepHours").DefaultValue = CInt(PrepHours.Text)
        SqlDataSource1.InsertParameters("prepMins").DefaultValue = CInt(PrepMins.Text)
        SqlDataSource1.InsertParameters("cookHours").DefaultValue = CInt(CookHours.Text)
        SqlDataSource1.InsertParameters("cookMins").DefaultValue = CInt(CookMins.Text)
        SqlDataSource1.InsertParameters("Servings").DefaultValue = CInt(Servings.Text)
        SqlDataSource1.InsertParameters("Calories").DefaultValue = CInt(Cals.Text)
        SqlDataSource1.InsertParameters("Fat").DefaultValue = CInt(Fat.Text)
        SqlDataSource1.InsertParameters("Chol").DefaultValue = CInt(Chol.Text)
        SqlDataSource1.InsertParameters("Carbo").DefaultValue = CInt(Carbo.Text)
        SqlDataSource1.InsertParameters("Ingredients").DefaultValue = Ingredients.Text
        SqlDataSource1.InsertParameters("S1").DefaultValue = {S1.Text + " "}
        SqlDataSource1.InsertParameters("S2").DefaultValue = {S2.Text + " "}
        SqlDataSource1.InsertParameters("S3").DefaultValue = {S3.Text + " "}
        SqlDataSource1.InsertParameters("S4").DefaultValue = {S4.Text + " "}
        SqlDataSource1.InsertParameters("S5").DefaultValue = {S5.Text + " "}
        SqlDataSource1.InsertParameters("Additional").DefaultValue = (Additional.Text + " ")
        SqlDataSource1.Insert()
        Response.Redirect("MainPage.aspx")
    End If
End Sub
```


Agree to purchase page

- Instantly redirects the users who have unlimited access
- Waits for users with “Single” recipe accounts to agree to the fee of \$1.
- Redirects the users who have yet to set up their account to the account management page

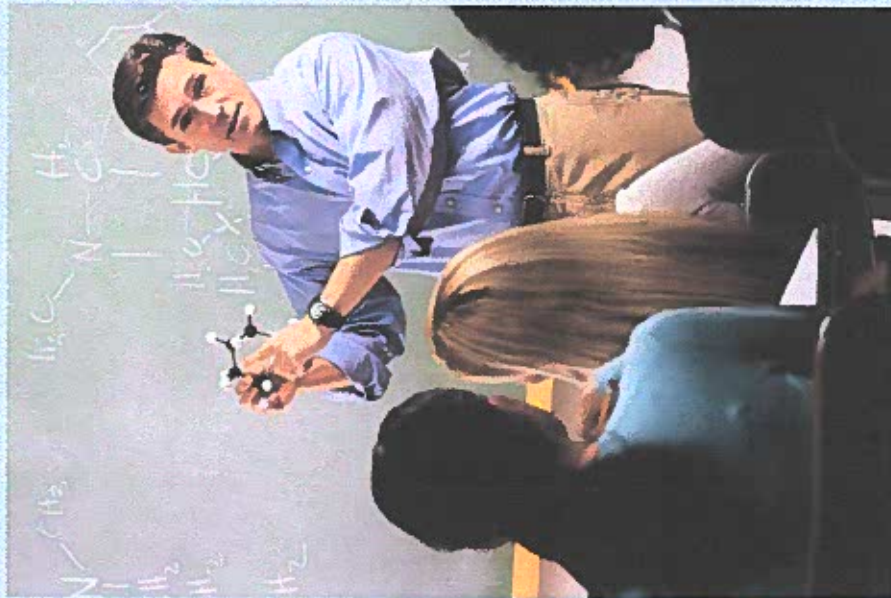
```
Protected Sub Page_Load(ByVal sender As Object, ByVal e As System.EventArgs) Handles Me.Load
    Dim reader As SqlDataReader
    SqlDataSource1.SelectCommand = "Select * FROM [AccountInfo] Where [AccountName] = '" & User.Identity.Name & "'"
    reader = SqlDataSource1.Select(DataSourceSelectArguments.Empty)
    If reader.Read() Then
        Dim aType As String = reader("AccountType")
        If aType = "Unlimited" Then
            Response.Redirect("ViewIT.aspx")
        End If
    Else
        Response.Redirect("ManageAccount.aspx")
    End If
End Sub
```


Three different Database Tables



- **AccountInfo**
 - Keeps track of each individual's account setup
- **Recipes**
 - The Database of all of the recipes
- **DollarsOwed**
 - A database that stores a list of the people who purchased a recipe individually so I know who to charge

Now to Demonstrate!



Program One

Program Two

Program Three

Program Four

Ethics Papers

Presentations